

$$x-y, \quad y-z, \quad z=5$$

(Ans.)

2012  
Q. Question: Discuss the Jacobi Iterative method for numerical solution of a system of linear equation.

Sol: The Jacobi Iterative Method is a direct method for the solution of a system of linear equations.

In this method we start from an approximation to the true solution and, if convergent, derive a sequence of closer approximations. We repeat the cycle of computations till the required accuracy is obtained.

Thus the computation depends on the accuracy required  
consider the system,

$$\left. \begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n &= b_2 \\ \dots &\dots \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n &= b_n \end{aligned} \right\} \quad (1)$$

In which the diagonal elements  $a_{ii}$  do not vanish

If it is not so then we can rearrange the equations so that this condition is satisfied.

Now the system (1) can be written as

$$\left. \begin{aligned} x_1 &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}}x_2 - \frac{a_{13}}{a_{11}}x_3 - \dots - \frac{a_{1n}}{a_{11}}x_n \\ x_2 &= \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}}x_1 - \frac{a_{23}}{a_{22}}x_3 - \dots - \frac{a_{2n}}{a_{22}}x_n \\ \dots &\dots \\ x_n &= \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}}x_1 - \frac{a_{n2}}{a_{nn}}x_2 - \dots - \frac{a_{n(n-1)}}{a_{nn}}x_{n-1} \end{aligned} \right\} \quad (2)$$

Let  $x_1^{(0)}, x_2^{(0)}$  be any first approximation to the unknowns  $x_1, x_2 = x_0$ . Putting these in the RHS of (a) we get a system of second approximations

$$\begin{aligned} x_1^{(1)} &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(0)} - \frac{a_{13}}{a_{11}} x_3^{(0)} \\ x_2^{(1)} &= \frac{b_2}{a_{21}} - \frac{a_{22}}{a_{21}} x_1^{(0)} - \frac{a_{23}}{a_{21}} x_3^{(0)} \end{aligned} \quad \left\{ \begin{array}{l} \text{RS} \end{array} \right.$$

$$x_3^{(1)} = \frac{b_3}{a_{31}} - \frac{a_{32}}{a_{31}} x_1^{(0)} - \frac{a_{33}}{a_{31}} x_2^{(0)} - \frac{a_{34}}{a_{31}} x_0^{(0)}$$

Again putting these in the right side of (a) we get the third approximations  $x_1^{(2)}, x_2^{(2)}, x_3^{(2)}$  in the same way if  $x_1^{(n)}, x_2^{(n)}, x_3^{(n)}$  are the  $n$ th approximations.

then the system of next approximation is given by

$$\begin{aligned} x_1^{(n+1)} &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(n)} - \frac{a_{13}}{a_{11}} x_3^{(n)} - \frac{a_{14}}{a_{11}} x_0^{(n)} \\ x_2^{(n+1)} &= \frac{b_2}{a_{21}} - \frac{a_{22}}{a_{21}} x_1^{(n)} - \frac{a_{23}}{a_{21}} x_3^{(n)} - \frac{a_{24}}{a_{21}} x_0^{(n)} \\ x_3^{(n+1)} &= \frac{b_3}{a_{31}} - \frac{a_{32}}{a_{31}} x_1^{(n)} - \frac{a_{33}}{a_{31}} x_2^{(n)} - \frac{a_{34}}{a_{31}} x_0^{(n)} \end{aligned}$$

Write the system (2) in matrix form we get

$$x = Bx + c$$

We can write the iteration formula (4) as

$$x^{(n+1)} = Bx^{(n)} + c$$

This method is known as Jacobi's Method of simultaneous displacement.

The sufficient condition for the convergence of this method is  $\|B\| < 1$

$x_j^{(1)} (j=1, 2, \dots, n)$  if every equation of the system (2) satisfies the condition that the sum of absolute value of the coefficients  $\frac{a_{ij}}{a_{ii}}$  at most equal to or in at least one equation less than unity, i.e. provided that

$$\sum_{j=1}^n \left| \frac{a_{ij}}{a_{ii}} \right| < 1, \quad (i=1, 2, \dots, n)$$

where the  $<$  sign should be valid in the case of at least one equation

The Jacobi method is called the method of simultaneous displacement.

Ques Problems: Using Jacobi's iterative method obtain iterative solution of the system.

$$10x - 5y - 22 = 3$$

$$4x - 10y + 32 = -3$$

$$x + 6y - 102 = -3$$

Sol<sup>n</sup> Given the system of solution

$$10x - 5y - 22 = 3$$

$$4x - 10y + 32 = -3$$

$$x + 6y - 102 = -3$$

----- (i)

equation (i) can be written as

$$10x = 3 + 5y + 22$$

$$\Rightarrow x = \frac{3}{10} + \frac{5}{10}y + \frac{2}{10}2$$

$$= 0.3 + 0.5y + 0.22 \quad \text{--- (ii)}$$

$$-10y = -3 - 4x - 32$$

$$\Rightarrow y = \frac{3}{10} + \frac{4}{10}x + \frac{3}{10}2 = 0.3 + 0.4x + 0.32 \quad \text{--- (iii)}$$

$$-102 = -3 - x - 6y$$

$$\Rightarrow x = \frac{3}{10} + \frac{1}{10}x + \frac{6}{10}y$$

$$= 0.3 + 0.1x + 0.6y$$

|         | x     | y     | z     |
|---------|-------|-------|-------|
| 1st it. | 0.3   | 0.3   | 0.3   |
| 2nd it. | 0.51  | 0.51  | 0.51  |
| 3rd it. | 0.655 | 0.657 | 0.657 |

(Ans.)

Ex Problems Use Jacobi iteration method to solve the following system.

$$2x + y + z = 5$$

$$3x + 5y + 2z = 15$$

$$2x + y + 4z = 8$$

Soln Given the system of equations

$$\left. \begin{aligned} 2x + y + z &= 5 \\ 3x + 5y + 2z &= 15 \\ 2x + y + 4z &= 8 \end{aligned} \right\} \text{--- (1)}$$

Equation (1) can be written as

$$2x = 5 - y - z$$

$$\Rightarrow x = \frac{5}{2} - \frac{1}{2}y - \frac{1}{2}z$$

The next approximation is

$$x^{(3)} = 0$$

$$z^{(3)} = 0$$

$$y^{(3)} = 0.7$$

The next approximation is

$$x^{(4)} = 0.15,$$

$$y^{(4)} = 0.3$$

$$z^{(4)} = 1.825$$

next app.

$$x^{(5)} = 0.0875$$

$$y^{(5)} = 0.98$$

$$z^{(5)} = 0.175$$

next app

$$x^{(6)} = 1.9225$$

$$y^{(6)} = 2.88$$

$$z^{(6)} = 1.71$$

next app.

$$x^{(7)} = 0.205$$

$$y^{(7)} = 1.625$$

$$z^{(7)} = 0.91875$$



Question: Describe Gauss-Seidal method for numerical solution of a system of linear equation

Soln The Gauss-Seidal method is a direct method for the solution of a system of numerical solution of a system of linear equation.

In this method we start from an approximation to the true solution and, if convergent, derive a sequence of closer approximations.

We repeat the cycle of computations till the required accuracy is obtained. Thus the computation depends on the accuracy required.

Let us consider the system of simultaneous linear

$$\left. \begin{array}{l} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m \end{array} \right\} \text{--- (I)}$$



suppose, the diagonal coefficients  $a_{ii}$  in (1) do not vanish. It is not so then we can rearrange the equation that this condition is satisfied.

Now, the system (1) can be written as

$$\begin{aligned} x_1 &= \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2 - \frac{a_{13}}{a_{11}} x_3 - \dots - \frac{a_{1n}}{a_{11}} x_n \\ x_2 &= \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1 - \frac{a_{23}}{a_{22}} x_3 - \dots - \frac{a_{2n}}{a_{22}} x_n \\ &\vdots \\ x_n &= \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1 - \frac{a_{n2}}{a_{nn}} x_2 - \dots - \frac{a_{n(n-1)}}{a_{nn}} x_{n-1} \end{aligned} \quad \text{--- (2)}$$

Let  $x_1^{(1)}, x_2^{(1)}, \dots, x_n^{(1)}$  be any first approximations to the unknowns  $x_1, x_2, \dots, x_n$ . Putting the first approximations  $(x_1^{(1)}, x_2^{(1)}, \dots, x_n^{(1)})$  into the R.H.S of the first equation of (2) and denote the result by  $x_1^{(2)}$ . Now we substitute  $(x_1^{(2)}, x_2^{(1)}, \dots, x_n^{(1)})$  in the second equation on the right of the system (2) and the result is denoted by  $x_2^{(2)}$ . We put  $(x_1^{(2)}, x_2^{(2)}, x_3^{(1)}, \dots, x_n^{(1)})$  in the third equation and denote the result by  $x_3^{(2)}$ . Continuing this process, we get  $(x_1^{(2)}, x_2^{(2)}, \dots, x_{n-1}^{(2)})$  in the last equation and denote the result by  $x_n^{(2)}$ .

Then we get,

$$x_1^{(2)} = \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(1)} - \frac{a_{13}}{a_{11}} x_3^{(1)} - \dots - \frac{a_{1n}}{a_{11}} x_n^{(1)}$$

$$x_2^{(2)} = \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1^{(2)} - \frac{a_{23}}{a_{22}} x_3^{(1)} - \dots - \frac{a_{2n}}{a_{22}} x_n^{(1)}$$

$$x_n^{(2)} = \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1^{(2)} - \frac{a_{n2}}{a_{nn}} x_2^{(2)} - \dots - \frac{a_{n(n-1)}}{a_{nn}} x_{n-1}^{(2)}$$

The first stage of iteration is completed. In the same way  $x_1^{(n)}, x_2^{(n)}, \dots, x_n^{(n)}$  are the approximation, then the system of next approximation is given by

$$x_1^{(n+1)} = \frac{b_1}{a_{11}} - \frac{a_{12}}{a_{11}} x_2^{(n)} - \frac{a_{13}}{a_{11}} x_3^{(n)} - \dots - \frac{a_{1n}}{a_{11}} x_n^{(n)}$$

$$x_2^{(n+1)} = \frac{b_2}{a_{22}} - \frac{a_{21}}{a_{22}} x_1^{(n+1)} - \frac{a_{23}}{a_{22}} x_3^{(n)} - \dots - \frac{a_{2n}}{a_{22}} x_n^{(n)}$$

$$x_n^{(n+1)} = \frac{b_n}{a_{nn}} - \frac{a_{n1}}{a_{nn}} x_1^{(n+1)} - \frac{a_{n2}}{a_{nn}} x_2^{(n+1)} - \dots - \frac{a_{n(n-1)}}{a_{nn}} x_{n-1}^{(n+1)}$$

In this method we use an improved component as soon as it is available.

For any choice of the first approximation  $x_j^{(1)}$  ( $j=1, 2, \dots, n$ ) the Gauss-Seidal methods converge if the following condition is satisfied by every equation of the system (2). The sum of the absolute value of the coefficients  $\frac{a_{ij}}{a_{ii}}$  is at most equal to, or in at least one equation less than unity, i.e. if

$$\sum_{\substack{j=1 \\ (j \neq i)}}^n \left| \frac{a_{ij}}{a_{ii}} \right| \leq 1, \quad (i=1, 2, \dots, n).$$

Where the sign  $<$  is valid in the case of at least one equation.

Gauss-Seidal method is called the method of successive displacements.

201. Problem: Find the system of equations

$$2x - 3y + 10z = 9$$

$$89x + 11y - 4z = 95$$

$$7x + 52y + 13z = 104$$

$$3x + 8y + 29z = 71$$

using Gauss-Seidal method  
upto three decimals, or obtain 2<sup>nd</sup> iterated solution  
of the system

Sol<sup>n</sup> Given equations are

$$\left. \begin{aligned} 89x + 11y - 4z &= 95 \\ 7x + 52y + 13z &= 104 \\ 3x + 8y + 29z &= 71 \end{aligned} \right\} \text{--- (1)}$$

The given system can be written as

$$89x = 95 - 11y + 4z$$

$$\Rightarrow x = \frac{95}{89} - \frac{11}{89}y + \frac{4}{89}z$$

$$= 1.145 - 0.123y + 0.048z$$

$$52y = 104 - 7x - 13z$$

$$\Rightarrow y = \frac{104}{52} - \frac{7}{52}x - \frac{13}{52}z$$

$$= 2 - 0.035x - 0.25z$$

$$29z = 71 - 3x - 8y$$

$$\Rightarrow z = 2.448 - 0.103x - 0.276y$$

$$\text{Now } \left| \frac{-11}{83} \right| + \left| \frac{4}{83} \right| < 1, \quad \left| \frac{-7}{52} \right| + \left| \frac{-13}{52} \right| < 1, \quad \left| \frac{-3}{29} \right| + \left| \frac{-8}{29} \right| < 1$$

Thus the system satisfy the convergence condition  $\|B\| < 1$ . starting with initial approximation

$x^{(1)} = 0, y^{(1)} = 0, z^{(1)} = 0$  we get,

$$x^{(2)} = 1.1445, \quad y^{(2)} = 1.846, \quad z^{(2)} = 1.8206$$

The next approximation is

$$x^{(3)} = \frac{95}{83} - \frac{11}{83} (1.846) + \frac{4}{83} (1.8206) \\ = 0.99$$

$$y^{(3)} = \frac{104}{52} - \frac{7}{52} (0.99) - \frac{13}{52} (1.8206) \\ = 1.412$$

$$z^{(3)} = \frac{71}{29} - \frac{3}{29} x^{(3)} - \frac{8}{29} y^{(3)} \\ = \frac{71}{29} - \frac{3}{29} (0.99) - \frac{8}{29} (1.412) = 1.96$$

$$x^{(4)} = \frac{95}{83} - \frac{11}{83} y^{(3)} + \frac{4}{83} (z^{(3)}) \\ = \frac{95}{83} - \frac{11}{83} (1.412) + \frac{4}{83} (1.96) \\ = 1.052$$

$$\begin{aligned}
 y^{(4)} &= \frac{104}{52} - \frac{7}{52} x^{(4)} - \frac{13}{52} z^{(3)} \\
 &= \frac{104}{52} - \frac{7}{52} (1.052) - \frac{13}{52} (1.96) \\
 &= 1.3684
 \end{aligned}$$

$$\begin{aligned}
 z^{(4)} &= \frac{71}{29} - \frac{3}{29} x^{(4)} - \frac{8}{29} y^{(4)} \\
 &= \frac{71}{29} - \frac{3}{29} (1.052) - \frac{8}{29} (1.3684) \\
 &= 1.96196
 \end{aligned}$$

$$\begin{aligned}
 x^{(5)} &= \frac{95}{83} - \frac{11}{83} y^{(4)} + \frac{4}{83} z^{(4)} \\
 &= \frac{95}{83} - \frac{11}{83} (1.3684) + \frac{4}{83} (1.96196) \\
 &= 1.05778
 \end{aligned}$$

$$\begin{aligned}
 y^{(5)} &= \frac{104}{52} - \frac{7}{52} x^{(5)} - \frac{13}{52} z^{(4)} \\
 &= \frac{104}{52} - \frac{7}{52} (1.05778) - \frac{13}{52} (1.96196) \\
 &= 1.36712
 \end{aligned}$$

$$\begin{aligned}
 z^{(5)} &= \frac{71}{29} - \frac{3}{29} x^{(5)} - \frac{8}{29} y^{(5)} \\
 &= \frac{71}{29} - \frac{3}{29} (1.05778) - \frac{8}{29} (1.36712) \\
 &= 1.96171
 \end{aligned}$$



Next app.

$$x^{(6)} = \frac{95}{83} - \frac{11}{83} y^{(5)} + \frac{4}{83} z^{(5)}$$

$$= \frac{95}{83} - \frac{11}{83} (1.36712) + \frac{4}{83} (1.96171)$$

$$= 1.0579$$

$$y^{(6)} = \frac{104}{52} - \frac{7}{52} x^{(6)} - \frac{13}{52} z^{(5)}$$

$$= \frac{104}{52} - \frac{7}{52} (1.0579) - \frac{13}{52} (1.96171)$$

$$= 1.3672$$

$$z^{(6)} = \frac{71}{29} - \frac{3}{29} x^{(6)} - \frac{8}{29} y^{(6)}$$

$$= \frac{71}{29} - \frac{3}{29} (1.0579) - \frac{8}{29} (1.3672)$$

$$= 1.96168$$

Since  $x^{(6)}$ ,  $y^{(6)}$  and  $z^{(6)}$  are sufficiently close  
 $x^{(5)}$ ,  $y^{(5)}$  and  $z^{(5)}$  respectively. so the values  
1.057, 1.367 and 1.961 be taken as the  
solution of linear equation.



Q.38 Problems Using the Gauss-Seidal iteration method obtain and iterated solution of the system

$$2x_1 + x_2 + x_3 = 5$$

$$3x_1 + 5x_2 + 2x_3 = 15$$

$$2x_1 + x_2 + 4x_3 = 8$$

sol<sup>n</sup> Given equations are

$$2x_1 + x_2 + x_3 = 5$$

$$3x_1 + 5x_2 + 2x_3 = 15$$

$$2x_1 + x_2 + 4x_3 = 8$$

----- (1)

We can write this system in the form

$$2x_1 = 5 - x_2 - x_3$$

$$\Rightarrow x_1 = \frac{5}{2} - \frac{1}{2}x_2 - \frac{1}{2}x_3$$

$$5x_2 = 15 - 3x_1 - 2x_3$$

$$\Rightarrow x_2 = \frac{15}{5} - \frac{3}{5}x_1 - \frac{2}{5}x_3$$

and

$$2x_3 = 8 - 2x_1 - x_2$$

$$\Rightarrow x_3 = \frac{8}{4} - \frac{2}{4}x_1 - \frac{1}{4}x_2$$

Now

$$|-\frac{1}{2}| + |-\frac{1}{2}| = 1, \quad |-\frac{3}{5}| + |-\frac{2}{5}| \leq 1, \quad |-\frac{2}{4}| + |-\frac{1}{4}| < 1$$

starting with initial approximation is

$$x_1^{(1)} = 0, x_2^{(1)} = 0, x_3^{(1)} = 0$$

next 1st iteration

$$x_1^{(2)} = \frac{5}{2}, \quad x_2^{(2)} = \frac{15}{5}, \quad x_3^{(2)} = \frac{8}{4}$$

and iteration  $x_2^{(2)} = 1.5$      $x_3^{(2)} = 0.375$

$$\begin{aligned} x_1^{(3)} &= \frac{5}{2} - \frac{1}{2} x_2^{(2)} - \frac{1}{2} x_3^{(2)} \\ &= \frac{5}{2} - \frac{1}{2} (1.5) - \frac{1}{2} (0.375) \\ &= 1.5625 \end{aligned}$$

$$\begin{aligned} x_2^{(3)} &= \frac{15}{5} - \frac{9}{5} x_1^{(3)} - \frac{2}{5} x_3^{(2)} \\ &= 3 - \frac{9}{5} (1.5625) - \frac{2}{5} (0.375) \\ &= 1.9125 \end{aligned}$$

$$\begin{aligned} x_3^{(3)} &= \frac{8}{4} - \frac{2}{4} x_1^{(3)} - \frac{1}{4} x_2^{(3)} \\ &= 2 - \frac{2}{4} (1.5625) - \frac{1}{4} (1.9125) \\ &= 0.7406 \end{aligned}$$

next approximation

$$\begin{aligned} x_1^{(4)} &= \frac{5}{2} - \frac{1}{2} x_2^{(3)} - \frac{1}{2} x_3^{(3)} \\ &= \frac{5}{2} - \frac{1}{2} (1.9125) - \frac{1}{2} (0.7406) \\ &= 1.17345 \end{aligned}$$

$$\begin{aligned}
 x_2^{(4)} &= \frac{15}{5} - \frac{3}{5} x_1^{(4)} - \frac{2}{5} x_3^{(3)} \\
 &= 3 - \frac{3}{5} (1.17345) - \frac{2}{5} (0.7406) \\
 &= 1.99969
 \end{aligned}$$

$$\begin{aligned}
 x_3^{(4)} &= \frac{8}{4} - \frac{3}{4} x_1^{(4)} - \frac{1}{4} x_2^{(4)} \\
 &= 2 - \frac{3}{4} (1.17345) - \frac{1}{4} (1.99969) \\
 &= 0.9134
 \end{aligned}$$

next app.

$$\begin{aligned}
 x_1^{(5)} &= \frac{5}{2} - \frac{1}{2} x_2^{(4)} - \frac{1}{2} x_3^{(4)} \\
 &= 2.5 - \frac{1}{2} (1.99969) - \frac{1}{2} (0.9134) \\
 &= 1.0435
 \end{aligned}$$

$$\begin{aligned}
 x_2^{(5)} &= \frac{15}{5} - \frac{3}{5} x_1^{(5)} - \frac{2}{5} x_3^{(4)} \\
 &= 3 - \frac{3}{5} (1.0435) - \frac{2}{5} (0.9134) \\
 &= 2.0085
 \end{aligned}$$

$$\begin{aligned}
 x_3^{(5)} &= \frac{8}{4} - \frac{3}{4} x_1^{(5)} - \frac{1}{4} x_2^{(5)} \\
 &= 2 - \frac{3}{4} (1.0435) - \frac{1}{4} (2.0085) \\
 &= 0.97612
 \end{aligned}$$

$$\begin{aligned}
 x_1^{(6)} &= \frac{5}{2} - \frac{1}{2} x_2^{(5)} - \frac{1}{2} x_3^{(5)} \\
 &= \frac{5}{2} - \frac{1}{2} (2.0085) - \frac{1}{2} (0.97612) \\
 &= 1.00769
 \end{aligned}$$

$$\begin{aligned}
 x_2^{(6)} &= \frac{15}{5} - \frac{3}{5} x_1^{(6)} - \frac{2}{5} x_3^{(5)} \\
 &= 3 - \frac{3}{5} (1.00769) - \frac{2}{5} (0.97612) \\
 &= 2.0049
 \end{aligned}$$

$$\begin{aligned}
 x_3^{(6)} &= \frac{8}{4} - \frac{3}{4} x_1^{(6)} - \frac{1}{4} x_2^{(6)} \\
 &= 2 - \frac{3}{4} (1.00769) - \frac{1}{4} (2.0049) \\
 &= 0.99499
 \end{aligned}$$

next app.

$$\begin{aligned}
 x_1^{(7)} &= \frac{5}{2} - \frac{1}{2} x_2^{(6)} - \frac{1}{2} x_3^{(6)} \\
 &= \frac{5}{2} - \frac{1}{2} (2.0049) - \frac{1}{2} (0.99499) \\
 &= 1.0000
 \end{aligned}$$

$$\begin{aligned}
 x_2^{(7)} &= \frac{15}{5} - \frac{3}{5} x_1^{(7)} - \frac{2}{5} x_3^{(6)} \\
 &= 3 - \frac{3}{5} (1.000) - \frac{2}{5} (0.99499) \\
 &= 2.0020
 \end{aligned}$$

$$\begin{aligned}
 x_3^{(7)} &= \frac{8}{4} - \frac{3}{4} x_1^{(7)} - \frac{1}{4} x_2^{(7)} = 2 - \frac{3}{4} (1.00) - \frac{1}{4} (2.0020) \\
 &= 0.9995
 \end{aligned}$$

$$x_1(7) = 1, \quad x_2(7) = 2 \quad \text{and} \quad x_3(7) = 0.99$$

Problems solve the following system of equations by Gauss-Seidel iterative method Ans:)

$$27x + 6y - z = 85,$$

$$6x + 15y + 2z = 72,$$

$$x + y + 54z = 110$$

Sol<sup>m</sup>

Given the system of equations

$$27x + 6y - z = 85$$

$$6x + 15y + 2z = 72$$

$$x + y + 54z = 110$$

— (i)

We can write this system of the form

$$27x = 85 - 6y + z$$

$$\Rightarrow x = \frac{85}{27} - \frac{6}{27}y + \frac{1}{27}z$$

$$15y = 72 - 6x - 2z$$

$$\Rightarrow y = \frac{72}{15} - \frac{6}{15}x - \frac{2}{15}z$$

— (ii)

and

$$54z = 110 - x - y$$

$$\therefore z = \frac{110}{54} - \frac{1}{54}x - \frac{1}{54}y$$

$$\text{Now, } \left| -\frac{6}{54} \right| + \left| \frac{1}{54} \right| < 1, \quad \left| -\frac{6}{15} \right| + \left| -\frac{2}{15} \right| < 1, \quad \left| -\frac{1}{54} \right| + \left| -\frac{1}{54} \right| < 1$$

Thus the convergency condition  $\|B\| < 1$  is satisfied

Now we start from an approximate value.

$$x^{(1)} = 0, y^{(1)} = 0, z^{(1)} = 0$$

1st approximation is

$$\begin{aligned} x^{(2)} &= 3.1481, \quad y^{(2)} = \frac{72}{15} - \frac{6}{15} x^{(1)} - \frac{2}{15} z^{(1)} \\ &= \frac{72}{15} - \frac{6}{15} (3.1481) - 0 \\ &= 4.8 - 1.25924 \\ &= 3.5408 \end{aligned}$$

$$\begin{aligned} z^{(2)} &= \frac{110}{54} - \frac{1}{54} x^{(2)} - \frac{1}{54} y^{(2)} \\ &= \frac{110}{54} - \frac{1}{54} (3.1481) - \frac{1}{54} (3.5408) \\ &= 1.9132 \end{aligned}$$

Next app.

$$\begin{aligned} x^{(3)} &= \frac{85}{27} - \frac{6}{27} y^{(2)} + \frac{1}{27} z^{(2)} \\ &= \frac{85}{27} - \left( \frac{6}{27} \right) (3.5408) + \frac{1}{27} (1.9132) \\ &= 2.4322 \end{aligned}$$

$$\begin{aligned} y^{(3)} &= \frac{72}{15} - \frac{6}{15} x^{(3)} - \frac{2}{15} z^{(2)} \\ &= \frac{72}{15} - \frac{6}{15} (2.4322) - \frac{2}{15} (1.9132) = 3.5720 \end{aligned}$$

$$z^{(3)} = \frac{110}{54} - \frac{1}{54} x^{(3)} - \frac{1}{54} y^{(3)}$$

$$= \frac{110}{54} - \frac{1}{54} (2.4322) - \frac{1}{54} (3.5720)$$

$$= 1.9258$$

Next app.

$$x^{(4)} = \frac{85}{27} - \frac{6}{27} y^{(3)} + \frac{1}{27} z^{(3)}$$

$$= \frac{85}{27} - \frac{6}{27} (3.5720) + \frac{1}{27} (1.9258)$$

$$= 2.4257$$

$$y^{(4)} = \frac{72}{15} - \frac{6}{15} x^{(4)} - \frac{2}{15} z^{(3)}$$

$$= \frac{72}{15} - \frac{6}{15} (2.4257) - \frac{2}{15} (1.9258)$$

$$= 3.5729$$

$$z^{(4)} = \frac{110}{54} - \frac{1}{54} x^{(4)} - \frac{1}{54} y^{(4)}$$

$$= \frac{110}{54} - \frac{1}{54} (2.4257) - \frac{1}{54} (3.5729)$$

$$= 1.92595$$

Since the value of  $x^{(4)}$ ,  $y^{(4)}$  and  $z^{(4)}$  are same to the value  $x^{(3)}$ ,  $y^{(3)}$  and  $z^{(3)}$  respectively. Therefore solution is  $x = 2.42$ ,  $y = 3.57$  and  $z = 1.92$  Ans.